

Introduction

Over-the-Air (OTA) Computing represents a paradigm shift in how distributed computations are executed across wireless networks. By exploiting the superposition property of the wireless Multiple-Access Channel (MAC), Over-the-Air (OTA) allows simultaneous transmissions from multiple sensing nodes to be aggregated in the air, directly yielding a desired nomographic function of the distributed data. This approach inherently couples communication and computation, breaking the separation principle to achieve unprecedented spectral efficiency and reduced latency. These properties are increasingly critical for rapidly scaling applications such as Federated Learning (FL), distributed consensus, and real-time sensor analytics.

0.a Background and Motivation

Traditional orthogonal access schemes, such as Time-Division Multiple Access (Time-Division Multiple Access (TDMA)), suffer from severe scalability bottlenecks because their resource requirements scale linearly with the number of transmitting nodes [5]. In computation-oriented networks where the receiver is interested only in a function of the data (e.g., the arithmetic mean) rather than the individual messages themselves, orthogonal separation proves highly suboptimal [3], [4]. OTA computation leverages analog signal superposition to compute these functions directly at the physical layer, achieving a computation rate that is theoretically independent of the network size [2]. However, achieving this theoretical efficiency in practice requires precise control over the transmitted waveforms and rigorous synchronization across spatially distributed transmitters.

0.b Problem Statement and Research Objectives

While the theoretical foundations of computation over MAC are well-established [1], [4], translating these scaling laws into practical realizations remains a formidable challenge. The primary focus of this thesis is to bridge the gap between idealized OTA theory and physical deployment. By systematically implementing OTA architectures across multiple abstraction layers—ranging from mathematical formulations and discrete-time MATLAB simulations to dynamical Simulink models and ultimately Software-Defined Radio (SDR) platforms—this work aims to quantify how different modeling fidelities capture the impact of physical channel impairments. Specifically, the objectives are to:

- Establish a rigorous ideal OTA baseline and validate its underlying physics.
- Systematically model and quantify the degradation caused by severe channel impairments, particularly multipath fractional delay and asynchronous Carrier Frequency Offsets (CFO).
- Evaluate established filtering and equalization techniques (such as squaring loops, Costas loops, and non-coherent envelope detection) to combat these uncoordinated impairments.
- Lay the groundwork for a robust Software Defined Radio (SDR) implementation using GNU Radio that addresses hardware-level synchronization limits.

0.c Thesis Contributions and Organization

This thesis contributes a comprehensive, cross-platform methodology for evaluating OTA computation under non-ideal physical conditions. The structure of the thesis is organized as follows:

- **Ideal Over-the-Air Computation Baseline:** Establishes the fundamental physical and mathematical models, exploring Double-Sideband Amplitude Modulation, matched filtering, and the theoretical Mean Squared Error (MSE) scaling under pure Additive White Gaussian Noise (AWGN).

- **Multipath Propagation, Fractional Delay, and Intersymbol Interference:** Dissects the vulnerability of OTA computation to multipath propagation and asynchronous CFO, providing detailed mathematical derivations, discrete-time simulations, and continuous-time Simulink block-level architectures that expose the mechanics of intersymbol interference and phase misalignment.
- **Experimental Implementation** (forthcoming): Transitions the modeled architectures into hardware, detailing the GNU Radio flowgraphs and the practical hurdles of SDR deployment.
- **Experimental Results and Discussion:** Provides a comparative analysis of the simulation models and equalization strategies, alongside an evaluation framework for the experimental SDR findings.
- **Conclusion and Future Work:** Synthesizes the findings and proposes advanced adaptive equalization schemes, such as Unscented Kalman Filtering, for future exploration.

References

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